Course: K-Means Clustering

Unsupervised Learning – Clustering

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Outline

What is clustering?

- Goal: group "similar" data points without labels.
- Clusters are **groups of points** that are close to each other.
- Applications:
  - customer segmentation,
  - image compression,
  - anomaly detection,
  - feature preprocessing for supervised ML.

Unsupervised learning

No ground truth $\rightarrow$ we search for hidden structure in the data.

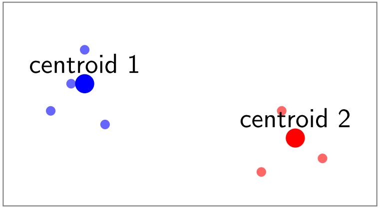
Intuition: group points by proximity

- We want to form K clusters.
- Each cluster has a **centroid** (center).
- Points are assigned to the nearest centroid (usually Euclidean distance).

Key idea

Minimize the total distance between points and their centroids.

Visual 1 — K-Means intuition



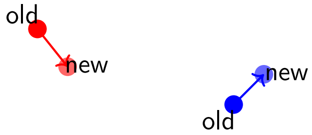
Two well-separated clusters → K-Means works perfectly here.

Algorithm steps

1. Choose K initial centroids (random or k-means++).
2. Assignment: each point $\rightarrow$ closest centroid.
3. Update: recompute each centroid (mean of assigned points).
4. Repeat steps 2–3 until convergence.

Convergence

The algorithm stops when centroid movement becomes negligible.



Iteration gradually stabilizes the centroid positions.

Objective function

Inertia (Within-Cluster Sum of Squares)

$$J = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

- x_i : point
- C_k : cluster k
- μ_k : centroid

Objective

Minimize $J \rightarrow$ compact, homogeneous clusters.

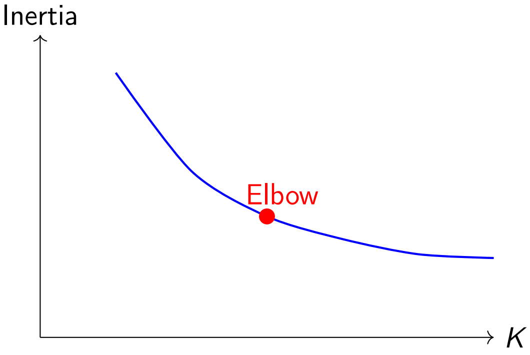
Elbow Method

- Compute inertia for several values of K.
- Plot inertia vs K.
- Find the point where improvement slows down.

Interpretation

Before the elbow $\rightarrow$ big gains. After the elbow $\rightarrow$ diminishing returns $\rightarrow$ good choice of K.

Visual 3 — Elbow method



The elbow suggests the optimal trade-off between simplicity and accuracy.

Silhouette Score

Definition

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

- $a(i)$: distance to its own cluster.
- $b(i)$: distance to the nearest other cluster.
- $s(i)$ ranges from -1 to 1.

Interpretation

Closer to 1 $\rightarrow$ better clustering.

Advantages of K-Means

- Very simple and intuitive.
- Extremely fast and scalable.
- Works well on spherical, well-separated clusters.

Use cases

Client segmentation, color quantization, fast partitioning.

Limitations of K-Means

- Requires choosing K.
- Assumes spherical clusters.
- Sensitive to outliers.
- Can converge to local minima.

Practical fix

Use `k-means++` initialization and several random starts.